

Comparative Analysis of ADAHESSIAN and First-Order Optimizers for CSI-Based Sign Language Recognition

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INTRODUCTION

- CSI data (amplitude & phase) is collected during gestures between a Wi-Fi Station (STA) and Access Point (AP).
- The raw data (3, 30, 200) for both amplitude and phase is preprocessed and reshaped into a (200, 60, 3) tensor.
- AdaHessian, a second-order optimizer, incorporates curvature info to potentially improve convergence and robustness.
- The goal is to compare second-order and first-order optimizers in terms of speed, noise resistance, and accuracy.

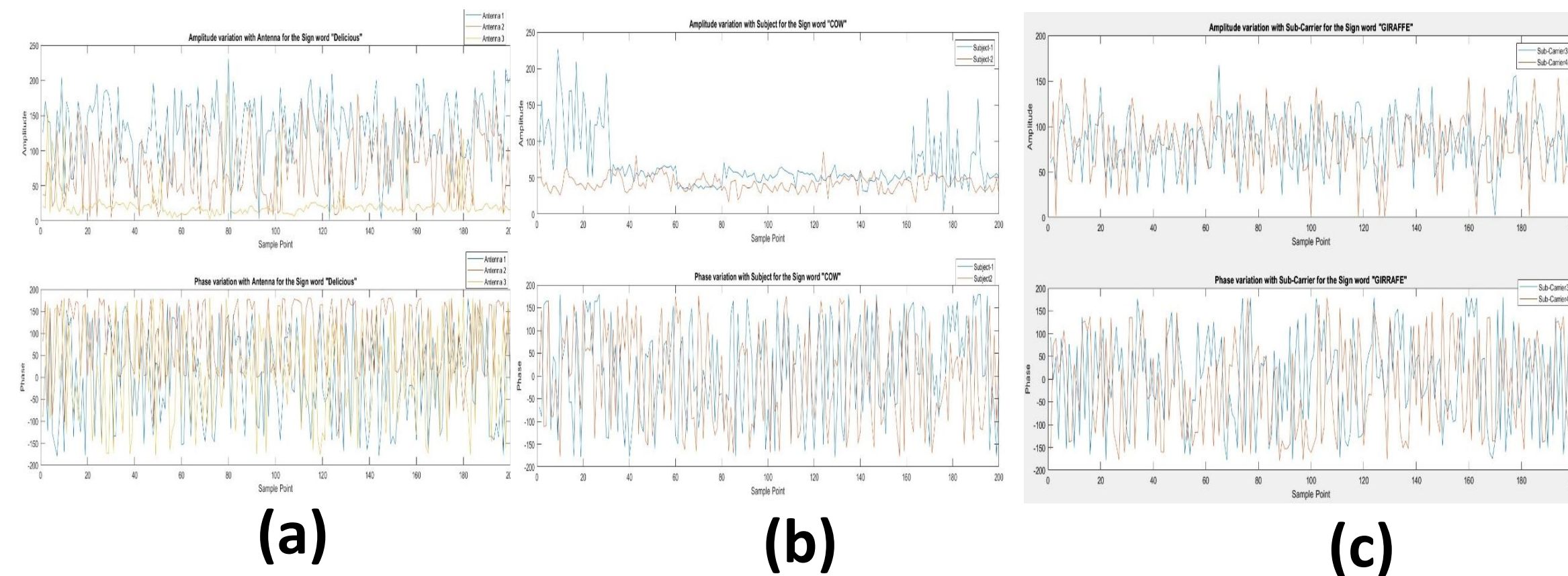


Figure 1: CSI Data Variation with (a) Antenna (b) Sub-Carrier (c) Subject

Problem Definition

- Descent Direction $\Delta w = \text{diag}(H)^{-k} g$, k is the hessian power
- Hessian Diagonal Estimation

$$D_t = \text{diag}(H_t) = \mathbb{E}[z \odot (H_t z)], \quad z \sim \text{Rademacher} \quad H_t z = \nabla^2 f(w_t) z = \frac{\partial}{\partial w} (g_t^\top z)$$

- Spatial Averaging

$$D_t^{(s)}[ib + j] = \frac{1}{b} \sum_{k=1}^b D_t[ib + k]$$

- Update Rule

$$w_t = w_{t-1} - \eta \cdot \frac{m_t}{v_t} \quad m_t = \frac{(1 - \beta_1) \sum_{i=1}^t \beta_1^{t-i} g_i}{1 - \beta_1^t} \quad v_t = (D_t)^k = \left(\sqrt{\frac{(1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} (D_t^{(s)}) \odot D_t^{(s)}}{1 - \beta_2^t}} \right)^k$$

Proposed Approach

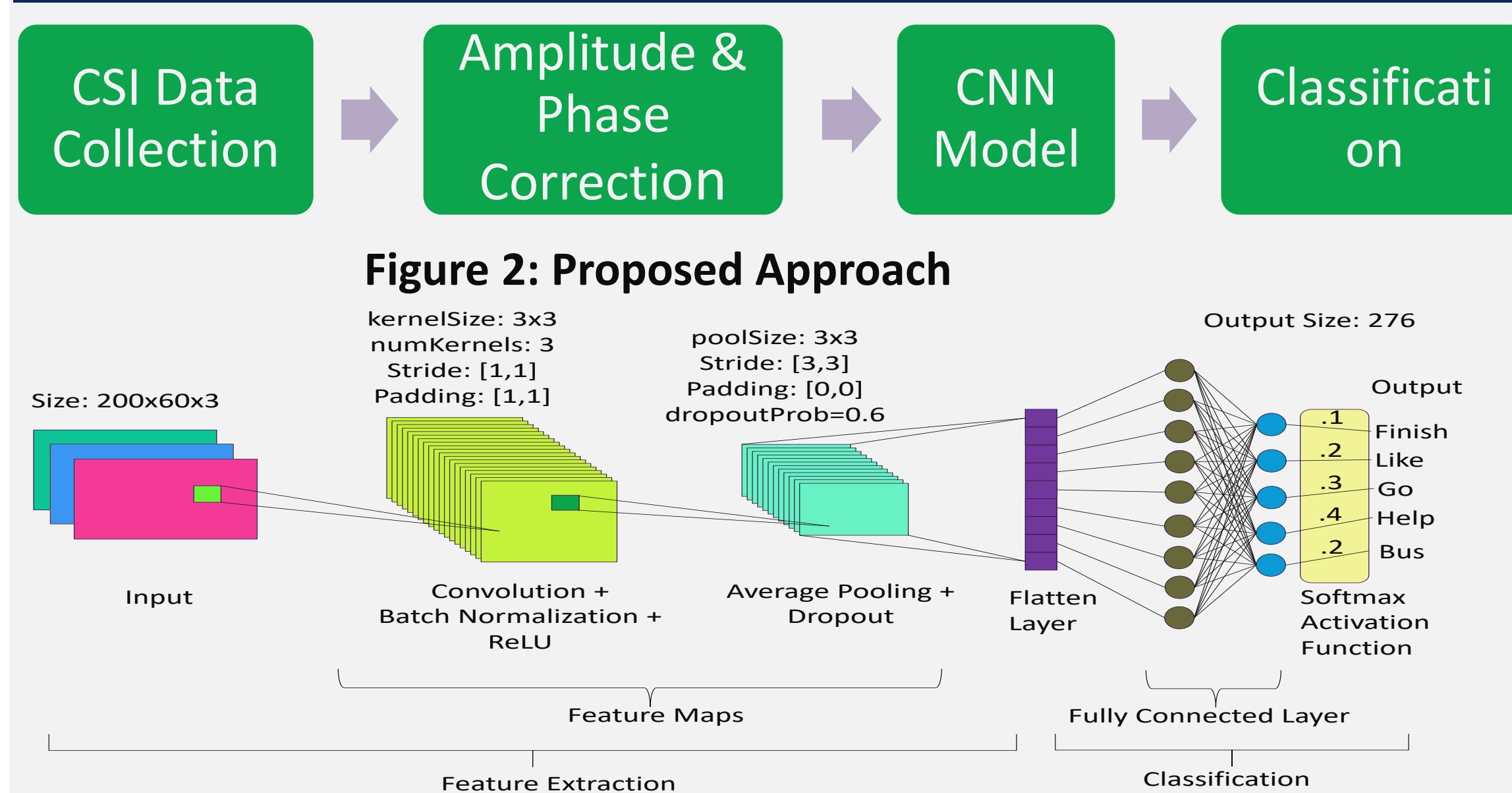
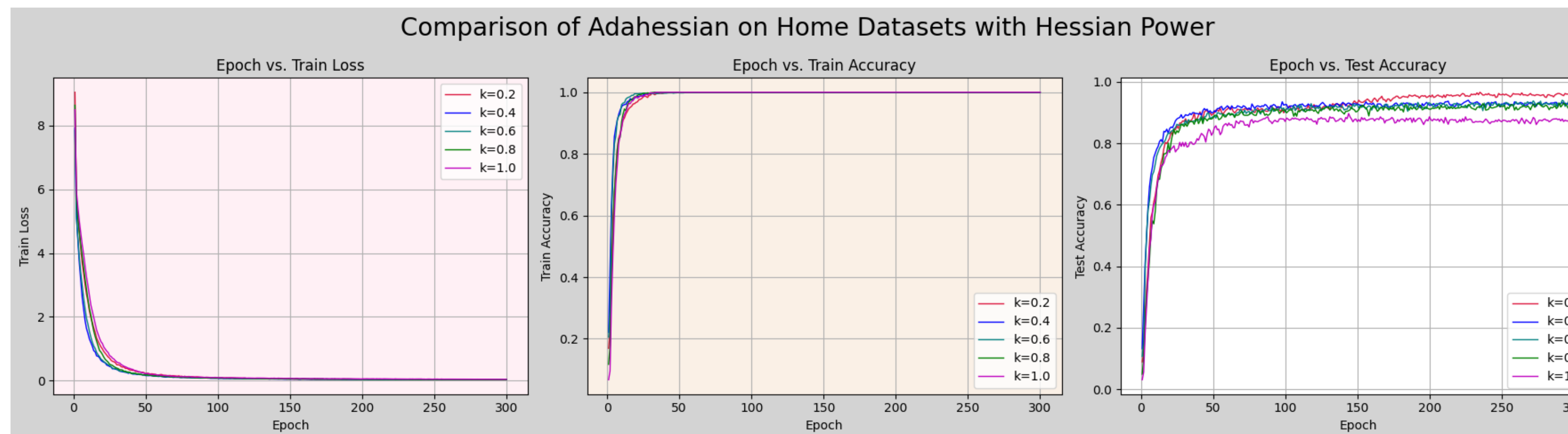
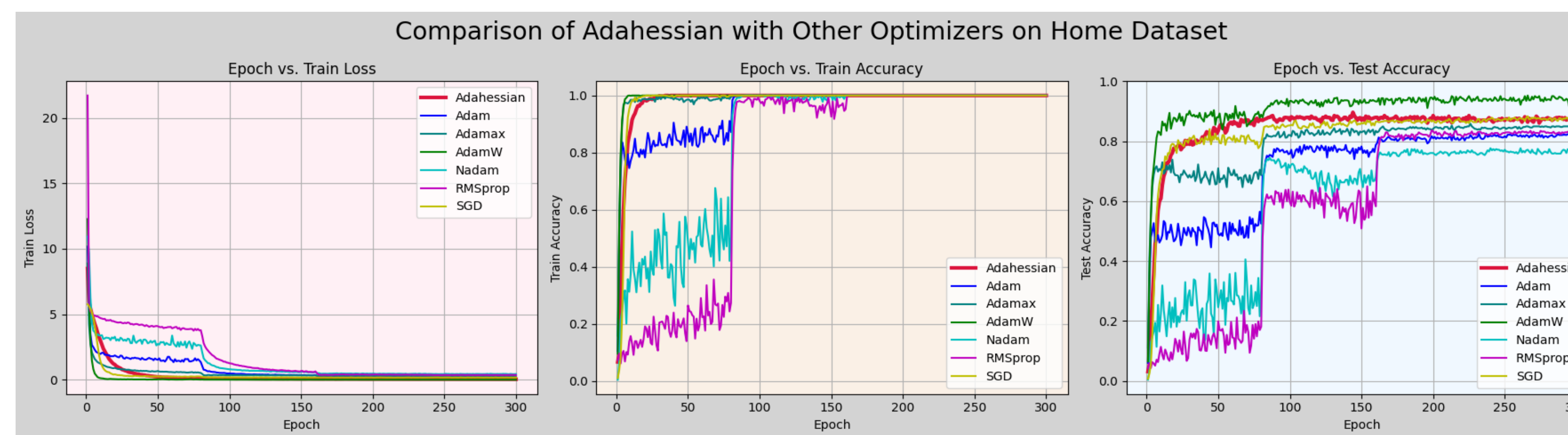


Figure 3: CNN Network Architecture

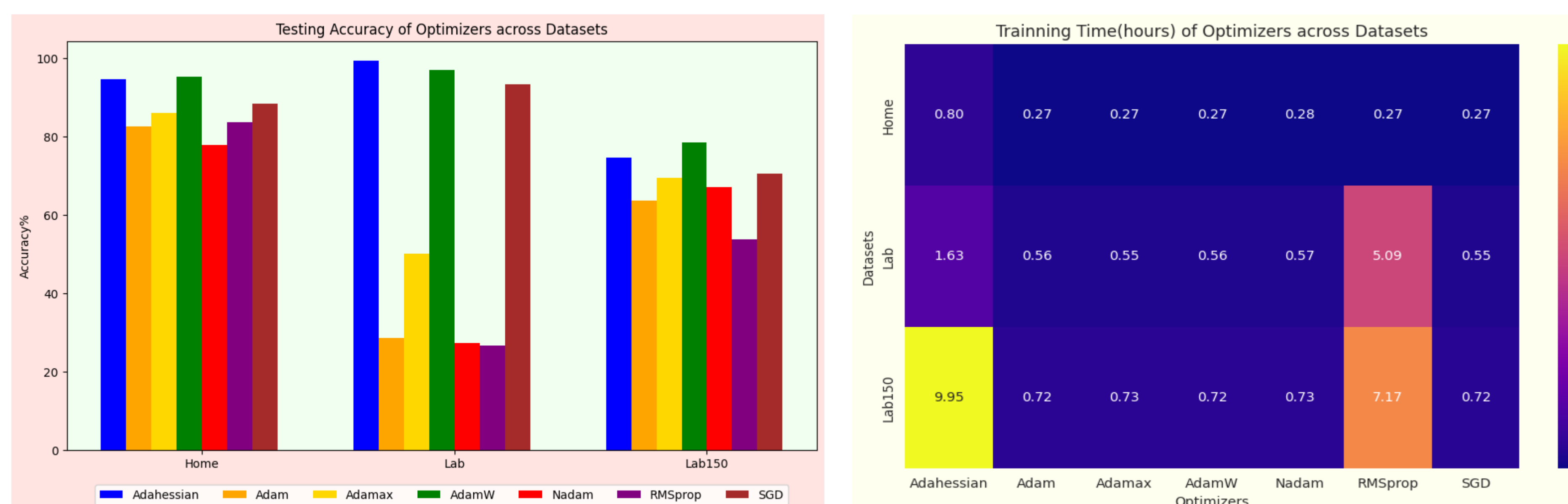
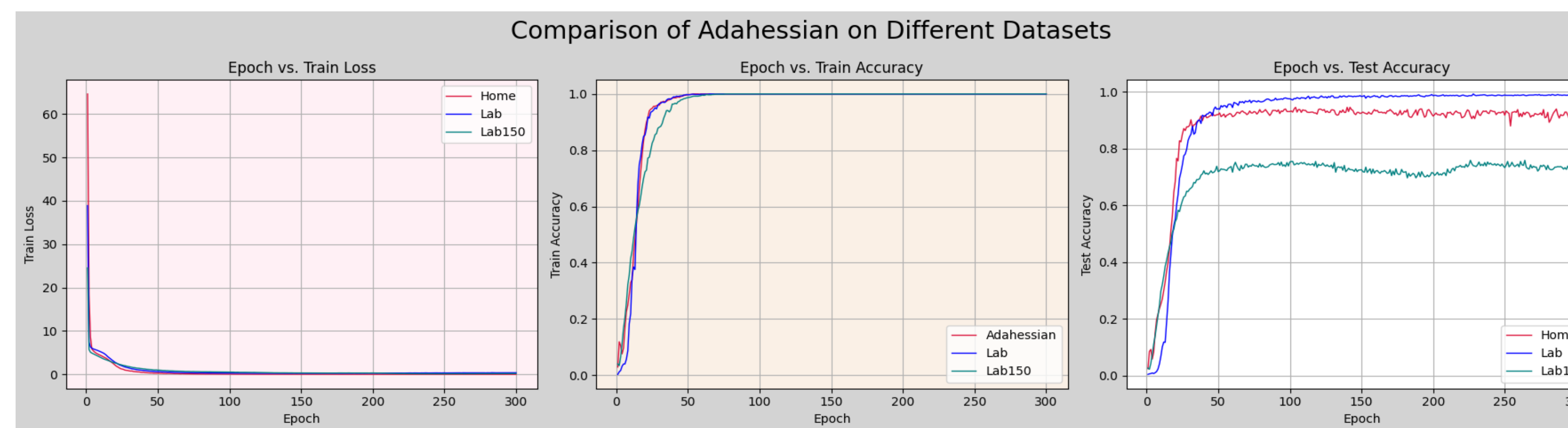
Experiments

Table 1: Dataset Used in the Experiment

Dataset	# Sign Labels	Repetitions Per Sign	# Instances
Home	276	10	2760
Lab	276	20	5520
Lab150	150	10	7500



$0 \leq k \leq 1$, $k=0 \rightarrow$ Gradient Descent, $k=1 \rightarrow$ Newton's Method



CONCLUSION

- AdaHessian outperforms others on the Lab dataset (+2.3% accuracy) and ranks second on Home and Lab150.
- Lab150 involves a multi-user setup, causing CSI data variations, leading to drop of accuracy(74.51%).
- Testing accuracy slightly varies with Hessian power(~7%).
- Used block size = 9, β_1 , β_2 as in Adam, and learning rate = 0.15 (no tuning).
- Future Work:
 - Apply delayed Hessian updates to reduce computational cost.
 - Tune β_1 , β_2 , weight decay, learning rate.
 - Test on cross-environment, cross-user settings.

REFERENCES

- Yongsan Ma, Gang Zhou, Shuangquan Wang, Hongyang Zhao, and Woosub Jung. 2018. SignFi: Sign Language Recognition Using WiFi. Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 2, 1, Article 23 (March 2018), 21pages. DOI: <https://doi.org/10.1145/3191755>
- Yao, Zhewei, Amir Gholami, Sheng Shen, Kurt Keutzer, and Michael W. Mahoney. 2021. ADAHESSIAN: An Adaptive Second Order Optimizer for Machine Learning. *Proceedings of the AAAI Conference on Artificial Intelligence* (AAAI 2021). DOI: <https://doi.org/10.48550/arXiv.2006.00719>

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